PVP 14

Code: CE1T6

I B.Tech - I Semester – Regular / Supplementary Examinations
December - 2016

BASIC MECHANICAL ENGINEERING
(CIVIL ENGINEERING)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1.

a) What is a mould and its role in casting?
b) What is the Principle of Gas Welding?
c) What is Pre ignition?
d) What is COP? Give an expression for COP of Refrigerator.
e) Draw a simple layout of a refrigeration system.
f) Explain the terms elasticity, factor of safety.
g) What is Resilience? Explain with a stress-strain curve.
h) Explain the terms :
   i) Strength ii) Hardness iii) Creep
i) Name the various types of gears.
j) List the various components of steam power plants.
k) List the various Alternate Energy Sources.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks. 3 x 16 = 48 M

2. a) Write the procedure for preparation of casting. 8 M

b) Explain the equipment of arc welding with a neat sketch. 8 M

3. a) Explain the working of four stroke petrol engine. 9 M

b) Differentiate between 2-stroke & four-stroke engines. 7 M

4. Explain the working of simple refrigeration system with a neat sketch. List out the applications of a refrigeration system. 16 M

5. a) Derive the relationship between young's modules, Bulk modules & Rigidity Modules. 8 M

b) What are the various electrical, magnetic & chemical properties of materials? Explain them in brief. 8 M

6. a) Differences between open belt & cross belt drives. Explain with diagram. Mention their applications. 8 M

b) Explain the layout of hydro electric power plant. 8 M

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(CIVIL ENGINEERING)

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PART – A

Answer ***all*** the questions. All questions carry equal marks
11x 2 = 22 M

1. a) What are the important components of an Internal Combustion Engine?
   b) In what aspects a 4-stroke C.I. Engine differs from that of an 4-stroke S.I. Engine.
   c) Define Coefficient of performance (COP).
   d) Draw the line diagram of vapor compression refrigeration system.
   e) Define stress, strain and Young's Modulus of a material.
   f) Define proof stress of a material and mention its significance.
   g) Write the expression for volumetric strain of a body in terms of its linear strain in orthogonal direction.
   h) Mention the types of belt drives.
   i) What do you mean by Foundry?
   j) What are the advantages of casting process?
   k) How power is developed by hydraulic turbine?

PART – B

Answer any ***THREE*** questions. All questions carry equal marks. 3 x 16 = 48 M

2. a) Explain the principles of gas welding and arc welding. 8 M

   b) Discuss various pattern allowances. 8 M

3. a) Explain why liquid fuels are mostly used in I.C. Engines. 4 M

   b) Describe the working of a 2-stroke Engine. Sketch its indicator diagram. 12 M

4. Draw the simple layout of summer air conditioning system and explain its working. 16 M

5. Explain in detail, the behavior of mild steel when subjected to a load test till failure. 16 M

6. a) Explain the working of a hydro electric power plant with a neat sketch. 8 M

   b) Write a short note on types of gears. 8 M

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I B.Tech - I Semester – Regular Examinations - January 2015

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Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks
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1. a) Define Core.
   b) List out various types of joints commonly used in welding.
   c) What is scavenging?
   d) What is the function of a carburettor?
   e) Define the term 'Air conditioning'.
   f) Define Ton of Refrigeration.
   g) What do you mean by Elastic Limit?
   h) Define Toughness.
   i) What is centrifugal tension in a belt ?
   j) Write down the working principle of tidal power plant.
   k) What is the use of surge tank in hydropower plants?

PART – B

Answer any ***THREE*** questions. All questions carry equal marks. 3 x 16 = 48 M

2. a) Name and describe the different properties of good moulding sands. 8 M

   b) Sketch the three types of gas welding flames and give differences between them. 8 M

3. a) Describe the working of four stroke SI engine. Illustrate using line diagrams. 8 M

   b) What are the advantages of a two stroke engine over a four stroke engine? 8 M

4. a) With the help of a neat schematic diagram, explain the working principle of a vapour compressor on refrigeration system. 8 M

   b) Draw the layout of a air-conditioning unit and explain its working. 8 M

5. a) Discuss briefly the various types of belts used for the transmission of power. 8 M

b) State the working principle and advantages of thermal power plant. 8 M

6. a) A rod 100 cm long and of 2 cm x 2 cm cross-section is subjected to a pull of 1000 kg force. If the modulus of elasticity of the materials 2.0 x 106 kg/cm^2, determine the elongation of the rod. 10 M

b) A load of 5 KN is to be raised with the help of a steel wire. Find the diameter of steel wire, if the maximum stress is not to exceed 100 MN/m^2. 6 M

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